PR1: Uninformed Search Algorithms

Implement Breadth First Search (BFS) algorithms for solving routefinding or maze navigation problems. Compare their performance based on path length, nodes explored and execution time.

PR2: Uninformed Search Algorithms

Depth First Search (DFS) algorithms for solving routefinding or maze navigation problems. Compare their performance based on path length, nodes explored and execution time.

PR3: Informed Search Algorithms

Implement Greedy Best First Search and A* Search algorithms for shortest path finding. Analyze optimality, completeness, and efficiency of both approaches.

PR4: Knowledge Representation and Expert System

Develop a rule-based expert system using propositional logic and implement Forward Chaining and Backward Chaining mechanisms for intelligent decision making.

PR5: Data Preprocessing and Exploratory Data Analysis

Download a standard dataset (Heart Disease, Diabetes or Titanic dataset) and perform:

- Data cleaning
- Missing value handling
- Feature scaling
- Encoding techniques
- Statistical analysis
- Data visualization using histograms and boxplots

Split the dataset into training and testing sets.

PR6: Regression Techniques

Implement Linear Regression and Polynomial Regression on a suitable dataset such as California Housing or House Price Prediction dataset.

Evaluate model performance using:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2 Score

Compare the results obtained by both techniques.

PR7: Classification using Logistic Regression and K-Nearest Neighbors

Perform data preprocessing and implement:

- Logistic Regression
- K-Nearest Neighbors (KNN)

Evaluate the models using:

- Accuracy
- Precision
- Recall
- Confusion Matrix
- F1-Score

Analyze and compare the results.

PR8:Classification using Decision Tree and Naïve Bayes Implement Decision Tree and Naïve Bayes classifiers on a suitable dataset. Compare classifier performance using standard evaluation metrics and identify the most suitable model for the selected dataset.
PR9: Clustering Techniques Implement: K-Means Clustering Hierarchical Clustering Use the Elbow Method and Dendrogram for determining the optimal number of clusters. Analyze the generated clusters using a real-world dataset.
PR10: Dimensionality Reduction using Principal Component Analysis Apply PCA on a high-dimensional dataset. Compare: Number of features before and after PCA Visualization of reduced dimensions Impact on classification performance
PR11: Advanced Machine Learning and Model Optimization Implement and compare: Support Vector Machine (SVM) Random Forest AdaBoost Perform: Cross Validation Hyperparameter Tuning using Grid Search Evaluate and compare model performance before and after optimization.